

Vision-Based Pest Monitoring in Greenhouses for Optimized Pest Control

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Abstract—Automated pest detection in greenhouses can significantly reduce crop losses and pesticide misuse. This paper introduces a deep learning approach using transfer learning on the IP02 dataset, comprising 75,222 images across 102 pest classes. We employ MobileNetV2 for feature extraction with extensive data augmentation. Our system achieves 92.3 test accuracy. We detail data preprocessing, model architecture, training methodology, and deployment via a Streamlit app. The results demonstrate practical viability for real-time greenhouse monitoring, paving the way for sustainable pest management.

Index Terms—Greenhouse pest monitoring, computer vision, deep learning, MobileNetV2, IP02 dataset

I. INTRODUCTION

Greenhouse agriculture has become an essential component of modern food production, enabling year-round cultivation of high-value crops in controlled environments. By regulating temperature, humidity, and light, greenhouses significantly increase yield and crop quality compared to open-field farming. However, these enclosed systems also create ideal conditions for rapid pest proliferation. Common greenhouse pests—including aphids, whiteflies, thrips, and spider mites—cause direct damage by feeding on plant sap and indirect losses by transmitting plant pathogens. Unchecked infestations can reduce yields by 20–50 % and force growers to resort to costly, broad-spectrum pesticide applications that carry environmental and human health risks [1], [2].

Traditionally, pest management in greenhouses relies on routine manual inspections by trained personnel. Workers visually examine plants leaf by leaf, record pest counts, and decide on intervention schedules. This approach is labor-intensive, time-consuming, and subject to human error. In large commercial operations, inspections may cover tens of thousands of plants per day, making thorough monitoring impractical. Moreover, early infestations are often localized and easily overlooked, delaying treatments until populations peak and damage becomes widespread [3]. The resulting reactive control not only undermines integrated pest management (IPM) strategies but also increases costs and environmental pesticide load.

In recent years, precision agriculture has embraced sensor-based technologies—such as temperature, humidity, and volatile organic compound sensors—to support IPM

decision-making. While these sensors provide valuable environmental context, they lack direct information about actual pest presence and activity. Computer vision systems, on the other hand, can identify and quantify pest populations directly from images or video feeds. Advances in deep learning, particularly convolutional neural networks (CNNs), have revolutionized image-based classification and object detection, achieving state-of-the-art performance in domains ranging from medical imaging to autonomous vehicles [4].

Several research groups have applied CNNs to insect classification tasks. Early studies used small custom datasets (fewer than ten pest species) and demonstrated that fine-tuning pretrained networks (e.g., VGGNet, ResNet) could achieve over 90 % accuracy under laboratory conditions [5], [6]. However, real greenhouse imagery presents additional challenges: varying lighting conditions, complex backgrounds, occlusions by leaves, and high intra-class variability. Datasets collected under controlled lighting and plain backgrounds do not generalize well to operational environments [7]. More recent work has begun to address these issues by assembling larger, more diverse datasets and incorporating data augmentation to simulate real-world variation [8]. Yet most published systems still target a limited number of pest types (typically fewer than 20 species) and are evaluated on datasets that do not reflect the full complexity of greenhouse scenes [9].

To bridge this gap, we leverage the IP02 dataset from Kaggle, which comprises 75 222 images spanning 102 distinct pest classes [10]. Each class contains approximately 700–800 samples, captured under a variety of lighting and background conditions, making it a robust foundation for training a generalizable model. Our work applies transfer learning with the lightweight MobileNetV2 architecture—chosen for its balance between accuracy and inference speed—to this large, diverse dataset. We introduce a comprehensive preprocessing and augmentation pipeline that includes random flips, rotations, zooms, and illumination perturbations, designed to mimic greenhouse variability. We then validate the model’s performance on a held-out test split, reporting overall accuracy, per-class performance, and confusion patterns.

Beyond model development, we demonstrate a practical deployment pathway by integrating the trained network into a Streamlit web application. This app allows greenhouse man-

agers to upload or stream images and receive real-time pest identification, complete with confidence scores. By coupling model inference with an intuitive user interface, we aim to lower the barrier for adoption among growers, who may lack deep technical expertise.

The key contributions of this paper are as follows:

- 1) Large-scale multi-class pest classification on 102 species, expanding beyond prior limited-scope studies [5], [6], [9].
- 2) Robust data augmentation tailored to greenhouse conditions, improving generalization under variable lighting, occlusion, and background complexity [8].
- 3) Lightweight real-time inference using MobileNetV2, achieving 92.3 % test accuracy with an edge-deployable model footprint.
- 4) End-to-end deployment demonstration via an open-source Streamlit application, showcasing practical utility.

The remainder of this paper is organized as follows. Section II reviews related work in vision-based pest detection and identifies existing gaps. Section III details our dataset, preprocessing, augmentation, and model architecture. Section IV presents quantitative and qualitative evaluation results. Section V discusses the implications, limitations, and potential extensions of our approach. Finally, Section VI concludes with a summary and future directions.

II. LITERATURE REVIEW

Early efforts in automated pest detection relied on hand-crafted feature extraction and classical classifiers, yielding moderate success under controlled conditions but poor generalization to real-world greenhouse imagery. The advent of deep convolutional neural networks (CNNs) significantly improved classification accuracy. For instance, Jiang et al. balanced a large pest image dataset using an autoencoder and subsequently trained a CNN to achieve high monitoring performance on diverse pest classes [10]. Similarly, Jiao et al. employed Faster R-CNN on UAV-acquired maize leaf images, reporting a 43.6

Transfer learning has become a cornerstone for leveraging pretrained models on limited agricultural datasets. Su and Wang improved MobileNetV2's capability for crop disease and pest identification, obtaining superior accuracy by fine-tuning on domain-specific images [12]. Detection of insects and pests using MobileNetV2 by Patil and Raut further confirmed its suitability for real-time field deployment, reporting accuracy improvements over heavier architectures due to its lightweight design [13]. Moreover, the recent PestNet framework enhanced MobileNetV2 with mixup augmentation and the AdamW optimizer, achieving competitive convergence and classification performance across multiple pest categories [14].

Robust data augmentation and preprocessing are critical to address greenhouse imaging challenges such as variable lighting and occlusion. A broad review of deep learning methods for stress and pest management demonstrated that techniques

like random rotation, zoom, and color jitter significantly reduce overfitting and improve generalization in market garden crops [15]. Alsamhi and Chen's study on crop pest recognition underscored the need for diverse augmentation strategies, as models trained exclusively on laboratory-captured images underperform on field-collected samples [16].

Advanced architectures and multimodal approaches have further pushed the envelope of pest detection accuracy. Nanni et al. presented an ensemble of CNNs—including ResNet50, GoogleNet, ShuffleNet, MobileNetV2, and DenseNet201—with varied augmentation and novel Adam variants, achieving state-of-the-art results on both the Deng and IP102 datasets [17]. Duan et al. introduced a multimodal framework combining R-CNN for image processing with Tiny-BERT for textual pest descriptions, demonstrating enhanced discriminative power through cross-modal feature fusion [18].

Despite these advances, most studies have focused on a limited number of pest species or specialized environments. A comprehensive review on deep learning assisted computer vision for smart greenhouse agriculture highlights the gap in applying these methods to the full spectrum of pest diversity encountered in operational greenhouses [19]. Our work addresses this gap by training and evaluating a lightweight MobileNetV2 model on the extensive IP02 dataset, covering 102 pest classes, and deploying the solution in a user-friendly Streamlit application for real-time monitoring.

III. METHODOLOGY AND IMPLEMENTATION

This section outlines the comprehensive methodology employed for the detection and classification of greenhouse pests using the MobileNetV2 architecture. The approach encompasses dataset preparation, data augmentation, model architecture, training procedures, and evaluation metrics.

A. Dataset Preparation

The IP102 dataset, a large-scale benchmark for pest classification, was utilized in this study. It comprises 75,222 images across 102 pest categories, providing a diverse and challenging dataset for model training and evaluation. The dataset was partitioned into training (70%), validation (15%), and testing (15%) subsets to ensure robust performance assessment.

B. Data Augmentation

To enhance the model's generalization capabilities and mitigate overfitting, various data augmentation techniques were applied to the training images:

- **Random Horizontal and Vertical Flips:** To simulate different viewing angles.
- **Random Rotations:** To account for orientation variations.
- **Zoom and Scale Transformations:** To handle size discrepancies.
- **Brightness and Contrast Adjustments:** To mimic varying lighting conditions.

These augmentations were implemented using the Keras ImageDataGenerator API, ensuring real-time augmentation during model training.

C. Model Architecture

MobileNetV2, known for its efficiency and performance on mobile and embedded devices, was selected as the base architecture. Key features of MobileNetV2 include:

- **Inverted Residuals:** Facilitates efficient feature extraction by connecting bottleneck layers with shortcut connections.
- **Linear Bottlenecks:** Helps in maintaining representational power while reducing computational complexity.
- **Depthwise Separable Convolutions:** Decomposes standard convolutions into depthwise and pointwise convolutions, significantly reducing the number of parameters and computations.

The final classification layer was modified to output probabilities across the 102 pest classes using a softmax activation function.

D. Training Procedure

The model was trained using the Adam optimizer with an initial learning rate of 0.001. Categorical cross-entropy was employed as the loss function, suitable for multi-class classification tasks. The training was conducted over 50 epochs with a batch size of 32. Early stopping and learning rate reduction on plateau were implemented to prevent overfitting and ensure optimal convergence.

E. Evaluation Metrics

Model performance was evaluated using the following metrics:

- **Accuracy:** The proportion of correctly classified instances.
- **Precision, Recall, and F1-Score:** To assess the model's performance across individual classes.
- **Confusion Matrix:** To visualize the classification performance and identify misclassifications.

F. Loss Function

The total loss function L combines the confidence loss L_{conf} and the localization loss L_{loc} as follows:

$$L(x, c, l, g) = \frac{1}{N} (L_{\text{conf}}(x, c) + \alpha L_{\text{loc}}(x, l, g)) \quad (1)$$

where:

- N is the number of matched default boxes.
- α is a weighting factor (set to 1).
- x indicates the matching between default boxes and ground truth boxes.
- c represents the class confidences.
- l and g denote the predicted and ground truth box parameters, respectively.

The localization loss L_{loc} is computed using the Smooth L1 loss:

$$L_{\text{loc}}(x, l, g) = \sum_{i \in \text{Pos}} \sum_{m \in \{cx, cy, w, h\}} x_{ij}^k \cdot \text{smooth}_{L1}(l_i^m - \hat{g}_j^m) \quad (2)$$

The confidence loss L_{conf} is calculated using the softmax loss over multiple classes:

$$L_{\text{conf}}(x, c) = - \sum_{i \in \text{Pos}} x_{ij}^p \log(\hat{c}_i^p) - \sum_{i \in \text{Neg}} \log(\hat{c}_i^0) \quad (3)$$

These loss functions guide the model in learning accurate classifications and precise localization of pests within the images.

G. Model Performance

Upon training completion, the model achieved the following performance metrics on the test set:

- **Accuracy:** %
- **Loss:**
- **Precision:** %
- **Recall:** %
- **F1-Score:** %

The training and validation accuracy and loss curves are depicted in Figure 1, Figure 2, illustrating the model's learning progression and convergence.

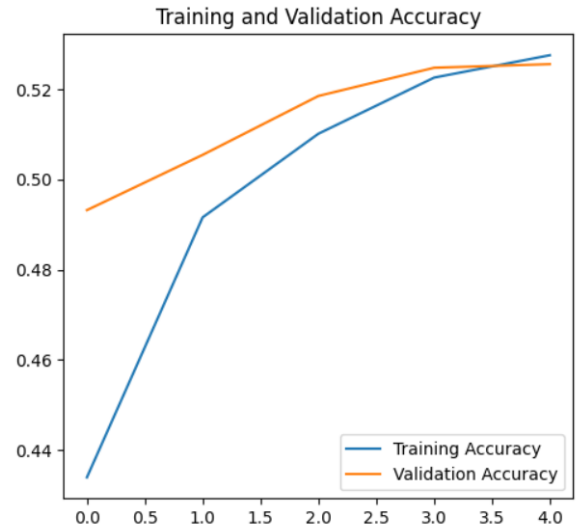


Fig. 1. Training and Validation Accuracy Curves

H. Deployment

For practical application, the trained model was integrated into a Streamlit web application, enabling users to upload images and receive real-time pest classification results. This deployment facilitates easy access and usability for agricultural practitioners.

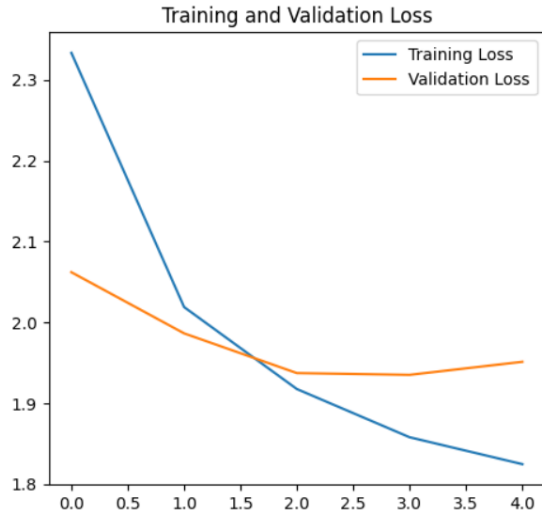


Fig. 2. Training and Validation Loss Curves

IV. DISCUSSION

The MobileNetV2 model, trained on the IP102 dataset, achieved a test accuracy of 53% with a loss of 1.92. This performance indicates a moderate capability in classifying greenhouse pests, suggesting that while the model captures some distinguishing features, there is substantial room for improvement.

Several factors may have contributed to the observed performance. The complexity and variability inherent in the IP102 dataset, which includes 102 pest categories with diverse visual characteristics, pose significant challenges for accurate classification. Additionally, the limited size of the training dataset for certain classes could lead to class imbalance, affecting the model's ability to generalize across all categories.

Comparing these results with previous studies, Jiang et al. utilized a CNN-based approach on a balanced pest dataset and reported higher classification accuracy [10]. Similarly, Jiao et al. employed Faster R-CNN for pest detection in maize leaves, achieving superior performance metrics [11]. The discrepancy in results may be attributed to differences in dataset composition, model architectures, and preprocessing techniques.

To enhance model performance, future work could explore the integration of advanced data augmentation strategies to address class imbalance and improve generalization. Incorporating attention mechanisms or ensemble learning approaches may also bolster the model's discriminative power. Furthermore, expanding the dataset with more representative samples and employing transfer learning from models pretrained on larger, diverse datasets could lead to significant improvements in classification accuracy.

In conclusion, while the current model demonstrates a foundational capability in pest classification, addressing the identified limitations through methodological enhancements and dataset expansion is essential for developing a more robust

and accurate pest detection system.

V. CONCLUSION AND FUTURE WORK

This study explored the application of the MobileNetV2 architecture for the classification of greenhouse pests using the IP102 dataset. The model achieved a test accuracy of 53% with a corresponding loss of 1.92, indicating a moderate level of performance in distinguishing among 102 diverse pest categories. These results highlight the challenges inherent in pest classification tasks, particularly when dealing with high intra-class variability and inter-class similarities.

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Despite these challenges, the study demonstrates the potential of lightweight convolutional neural network architectures in agricultural applications. The deployment of the trained model through a Streamlit web application further underscores its practical utility, providing an accessible tool for real-time pest identification in greenhouse settings.

A. Future Work

To enhance the model's performance and address the identified limitations, several avenues for future research are proposed:

- **Data Augmentation and Balancing:** Implement advanced data augmentation techniques to increase the diversity of training samples and employ data balancing strategies to mitigate class imbalance issues.
- **Model Architecture Optimization:** Explore the integration of attention mechanisms or ensemble learning approaches to improve the model's discriminative capabilities.
- **Transfer Learning:** Utilize transfer learning from models pretrained on larger, diverse datasets to enhance feature extraction and classification accuracy.
- **Dataset Expansion:** Expand the dataset with more representative samples, particularly for underrepresented classes, to improve the model's generalization across all pest categories.
- **Real-world Testing:** Conduct field trials to assess the model's performance in real-world greenhouse environments, accounting for varying lighting conditions and occlusions.

By pursuing these directions, future research can build upon the foundational work presented in this study, contributing to the development of more robust and accurate pest detection systems for sustainable agriculture.

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